

WEEK-7 PRACTICAL — Node.js User Service [Vikram Immella, 2520090118]

Step – 1 Create a User — POST

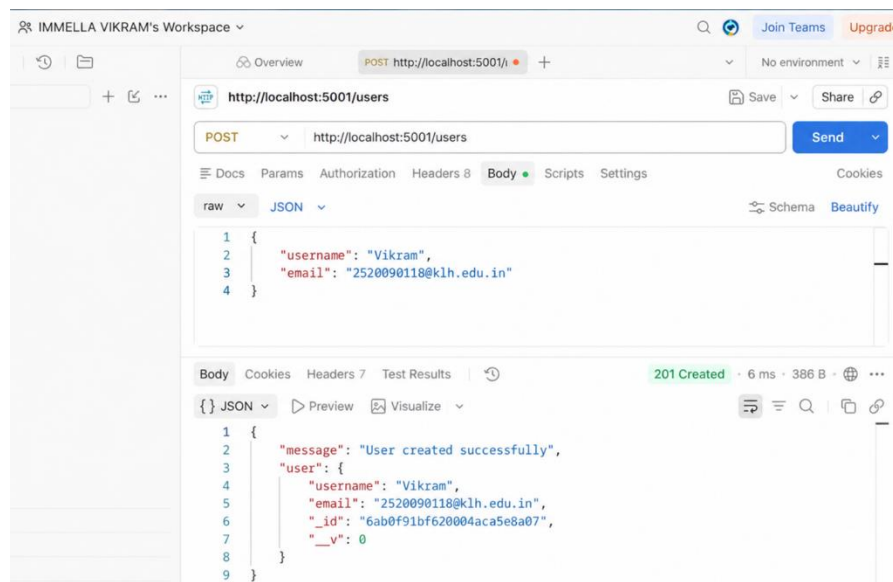
```
Windows PowerShell
-----
dar--l      21-09-2026    14:48      controllers
dar--l      21-09-2026    14:48      node_modules
dar--l      21-09-2026    14:48      routes
-a---l      21-09-2026    14:44      39264 package-lock.
-a---l      21-09-2026    14:44      json
-a---l      21-09-2026    14:44      349 package.json

PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\n
odememo\user-service> node app.js
User service running on port 5001
MongoDB connected

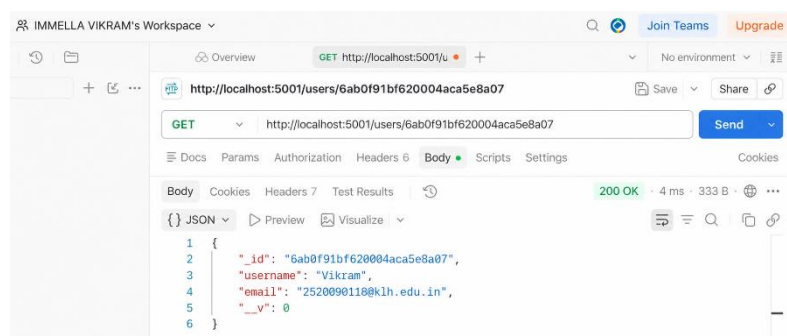
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\n
odememo\user-service> Invoke-RestMethod -Uri "http://localhost:5
001/users" -Method POST -ContentType "application/json" -Body '{
"username":"Vikram", "email": "2520090118@klh.edu.in", "profile":
{}}'

message          user
-----
User created successfully @(username=Vikram; email=2520090118...
```

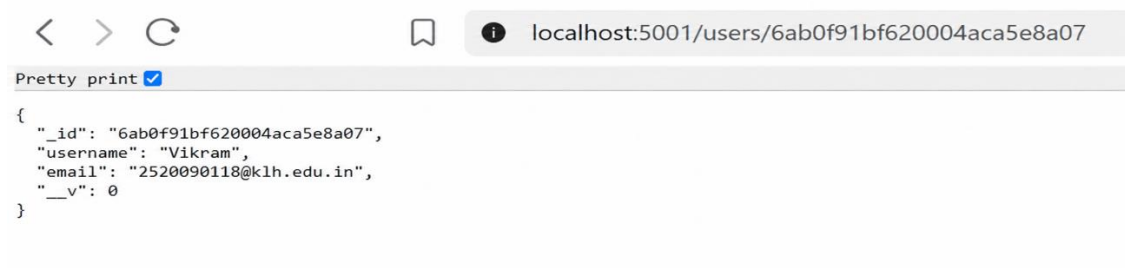
POST in mongosh:



Step 2: Get User — GET



Output on browser by id:

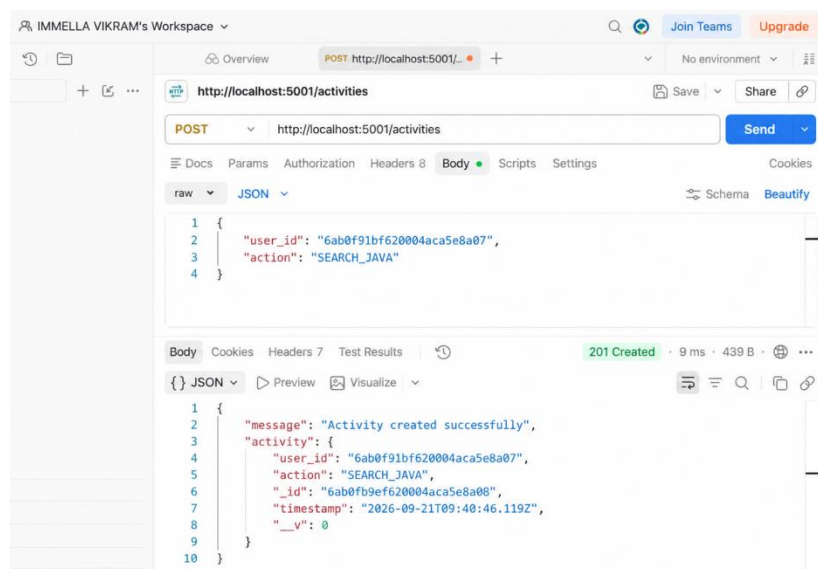


Step 3: Create Activity — POST

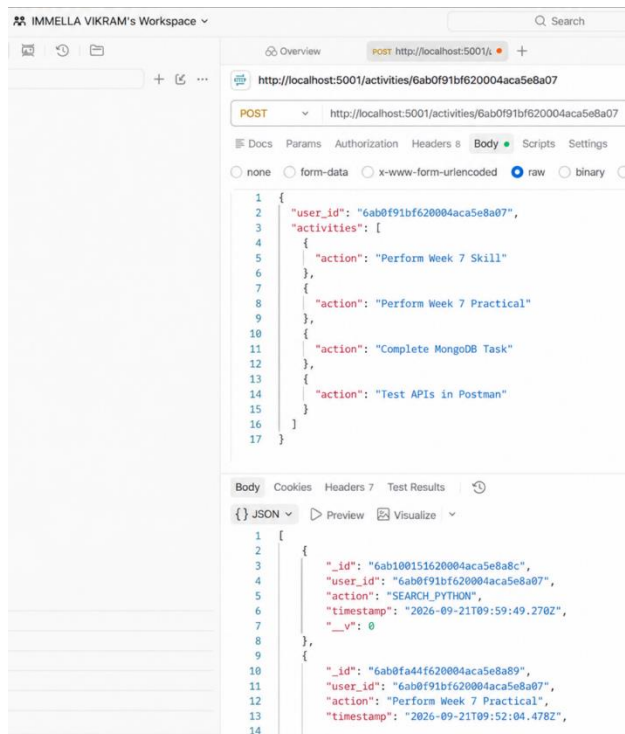
```
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\nodememo\user-service> Invoke-WebRequest -Uri "http://localhost:5001/activities" -Method POST -ContentType "application/json" -Body '{"user_id":"6ab0f91bf620004aca5e8a07","action":"SEARCH_PYTHON"}'
```

message activity
Activity created successfully @{user_id=6ab0f91bf620004aca5e...

In mongosh:



Step 4: Add More Activities



```

PS C:\Users\DELL\OneDrive\Documents\S DBEAD00\Practical\Week 7\nodememo\user-service> Invoke-WebRequest -Uri "http://localhost:5001/activities/6ab0f91bf620004aca5e8a07" -Method GET

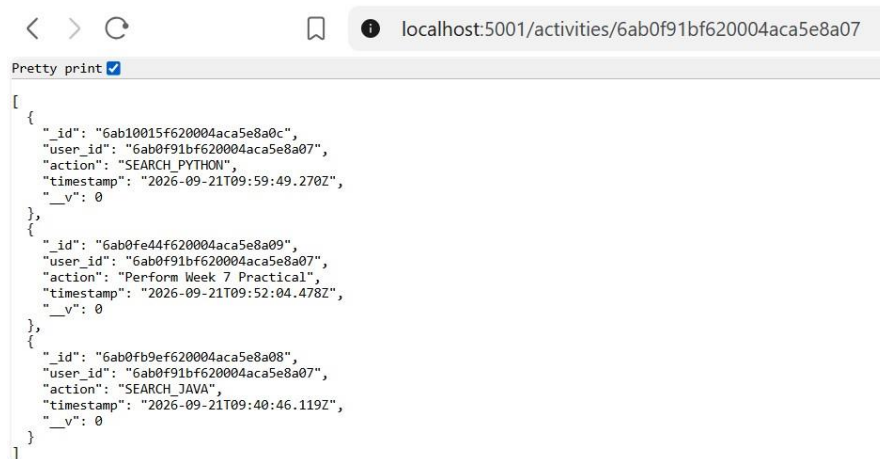
_id      : 6ab10015f620004aca5e8a0c
user_id   : 6ab0f91bf620004aca5e8a07
action    : SEARCH_PYTHON
timestamp : 2026-09-21T09:59:49.270Z
__v       : 0

_id      : 6ab0fe44f620004aca5e8a09
user_id   : 6ab0f91bf620004aca5e8a07
action    : Perform Week 7 Practical
timestamp : 2026-09-21T09:52:04.478Z
__v       : 0

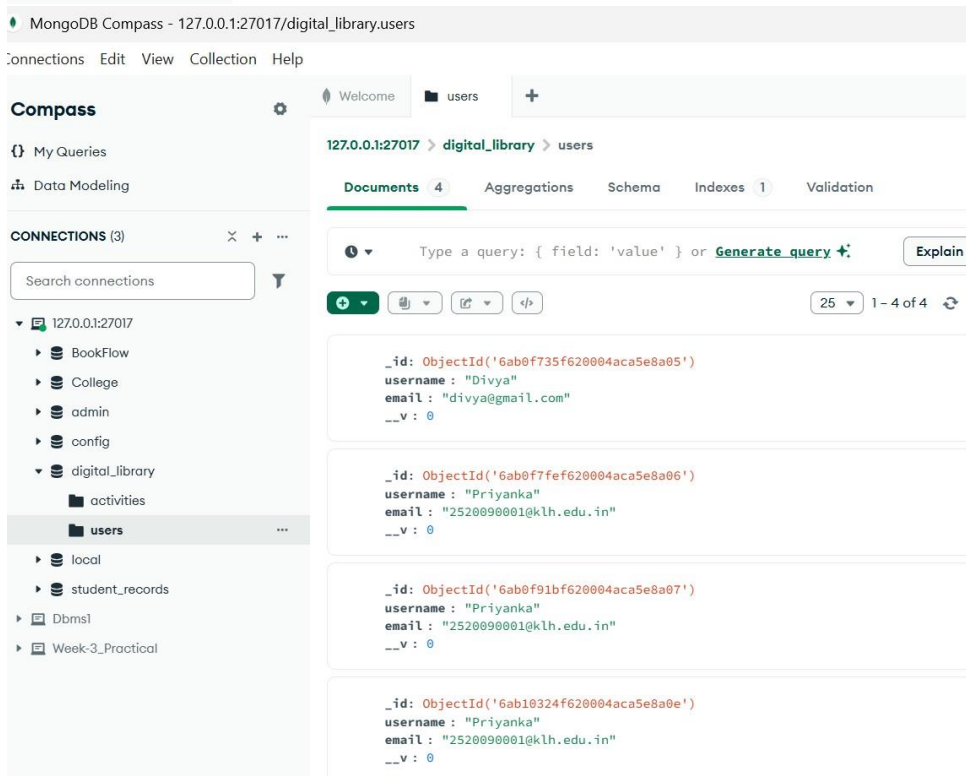
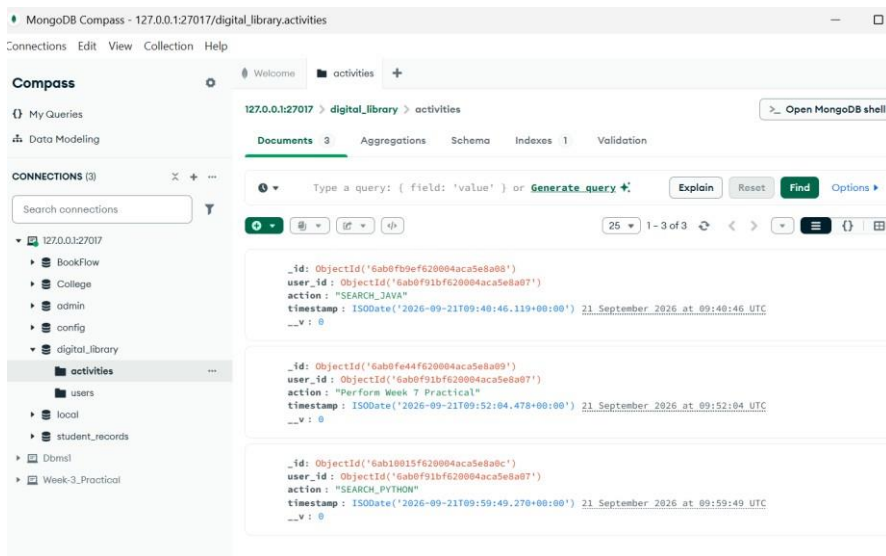
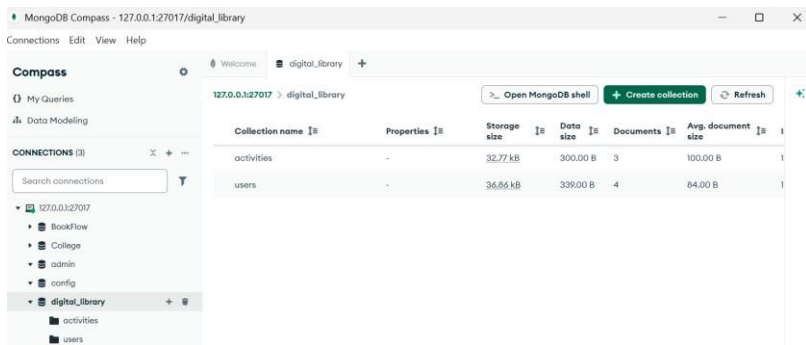
_id      : 6ab0fb9ef620004aca5e8a08
user_id   : 6ab0f91bf620004aca5e8a07
action    : SEARCH_JAVA
timestamp : 2026-09-21T09:40:46.119Z
__v       : 0

```

Output on browser:



MongoDB connection:



Folder Structure:

Week 7

```

|__nodememo
  |_user-service
    |
    ├── controllers
    |   ├── activityController.js
    |   └── userController.js
    |
    ├── routes
    |   ├── activityRoutes.js
    |   └── userRoutes.js
    |
    ├── app.js
    ├── db.js
    ├── package.json
    └── .env

```

db.js

```

const mongoose = require("mongoose");
const MONGO_URI =
process.env.MONGO_URI ||
  "mongodb://127.0.0.1:27017/digital_library";
mongoose.connect(MONGO_URI)
  .then(() => console.log("MongoDB connected"))
  .catch(err => console.error("MongoDB connection error:", err));
const userSchema = new
mongoose.Schema({      username: {
type: String,          required: true
},      email: {
type: String,
required: true
},      profile: {      type:
mongoose.Schema.Types.Mixed,
default: {}
}
}); const activitySchema = new
mongoose.Schema({      user_id: {
type: mongoose.Schema.Types.ObjectId,
required: true,        ref: "User"

```

```

    },    action: {
type: String,
required: true
    },    timestamp: {
type: Date,
default: Date.now
    }
}); const User = mongoose.model("User",
userSchema);
const Activity = mongoose.model("Activity", activitySchema);

module.exports = { User, Activity };

```

controllers/userController.js

```

const { User } = require("../db");
const createUser = async (req, res) => {    try {
const { username, email, profile } = req.body;
    const user = await
User.create({                username,
email,                profile
    });    res.status(201).json({
message: "User created successfully",
user    });    } catch (error) {
res.status(500).json({                message:
"Error creating user",                error:
error.message
    });
    }
}; const getUser = async (req, res) => {    try {
const user = await User.findById(req.params.id);
    if
(!user)
        return res.status(404).json({
message: "User not found"
    });
    res.json(user);    } catch
(error) {    res.status(500).json({
message: "Error getting user",
error: error.message
    });
    }
}; module.exports = { createUser,
getUser };

```

controllers/activityController.js

```

const mongoose = require("mongoose"); const
{ Activity } = require("../db");
const createActivity = async (req, res) => {
try {      const { user_id, action } =
req.body;
      if
(!mongoose.Types.ObjectId.isValid(user_id))
return res.status(400).json({      message:
"Invalid user_id"
});
      const activity = await
Activity.create({      user_id,
action
});      res.status(201).json({
message: "Activity created successfully",
activity      });      } catch (error) {
res.status(500).json({      message: "Error
creating activity",      error: error.message
});
    }
};
const getActivities = async (req, res) => {
try {      const { user_id } =
req.params;
      if
(!mongoose.Types.ObjectId.isValid(user_id))
return res.status(400).json({      message:
"Invalid user_id"
});

      const activities = await Activity
.find({ user_id })
      .sort({ timestamp: -1 });
      res.json(activities);      } catch
(error) {      res.status(500).json({
message: "Error getting activities",
error: error.message
});
    }
}; module.exports = { createActivity,
getActivities };
routes/userRoutes.js
const express =
require("express");
const router =
express.Router();
const {
createUser,
getUser
} = require("../controllers/userController");

```

```
    router.post("/users",
createUser);
router.get("/users/:id", getUser);
```

```
module.exports = router;
```

routes/activityRoutes.js

```
const express =
require("express");
const router =
express.Router();
const {
createActivity,
getActivities
} = require("../controllers/activityController");
```

```
router.post("/activities", createActivity);
router.get("/activities/:user_id", getActivities);
module.exports =
router;
```

```
app.js const express =
require("express");
const app =
express();
```

```
require("./db");
app.use(express.json());
const activityRoutes =
require("./routes/activit
yRoutes"); const
userRoutes =
require("./routes/userRou
tes");
app.use("/",
activityRoutes);
app.use("/", userRoutes);
app.listen(5001, () => { console.log("User service
running on port 5001");
});
```